

Specification: Strategies for the automated identification of service interfaces during the migration from monoliths to microservices

Alexander Ley
Computer Science Department
University of applied science Bonn-Rhein-Sieg
Sankt Augustin, Germany
s6alleyy@outlook.de

I. CONTEXT AND MOTIVATION

In Software Engineering of business applications, the architecture and organization of software components are playing a pivotal rule to enable desired features like *scalability*, *maintainability* and *deployability*.

As software systems become more and more complex these features are more and more less achieved due to the predominant implementation of *Monolithical Architectures* [1].

A. Monolithical Architectures

Traditionally, teams in Software Engineering of business applications are divided along technical capabilities.

According to Conway's Law, which is quoted in Quote 1, the technically oriented team decomposition yields to the technically oriented decomposition of the software output, which is illustrated in figure Fig. 1.

"Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure."

Quote 1. Conway's Law quoted by Melvin Conway, 1968 [2]

Monolithical Architectures decompose the software according to horizontal and technical oriented layers. Between these layers are often lots of dependencies resulting in a loss of flexibility whenever the software has to be adapted hindering agile software development and maintenance.

As illustrated in figure Fig. 3 monoliths are traditionally built as a single unit which can only be deployed and scaled as a whole. Therefore the deployability and scalability using a Monolithical Architecture is limited. [1]

As the requirements towards the software are evolving there is a need for software to be locally, flexible, adaptable and deployable without influencing other parts of software.

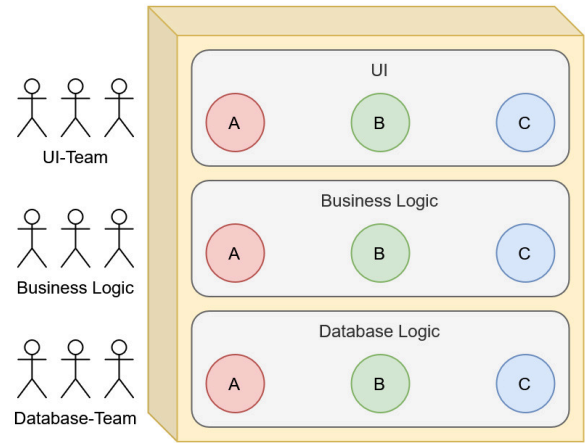


Fig. 1. Illustration of a business application built as a monolith. Original illustration from C. Nockemann [3].

B. Microservice Architecture

Due to the downsides of Monolithical Architectures researchers have reached out to other software paradigms [4].

Microservice Architectures seek to overcome the shortcomings of monoliths and have therefore gained significant attraction [5]. The core idea behind this novel kind of software organization is to decompose the software vertically along the enterprise business capabilities as illustrated in Fig. 2.

J. Lewis and M. Fowler [2] described the *Microservice Architecture* as follows:

"The microservice architectural style is an approach to developing a single application as a suite of small services, each running in its own process and communicating with lightweight mechanisms, often an HTTP resource API [today often a RESTful API [5]]. These services are built around business capabilities and independently deployable by fully automated deployment machinery. There is a bare minimum of centralized management of these services, which may be written in different programming languages and use different data storage technologies."

Microservices are typically build in a way that they encapsulate one single business capability as shown in Fig. 2 which can encompass an own database, own User-Interface (UI) and business logic seeking of being as independent as possible to other Microservices. Therefore they can be maintained, adapted and deployed individually which satisfy more companies desire for agile software development.

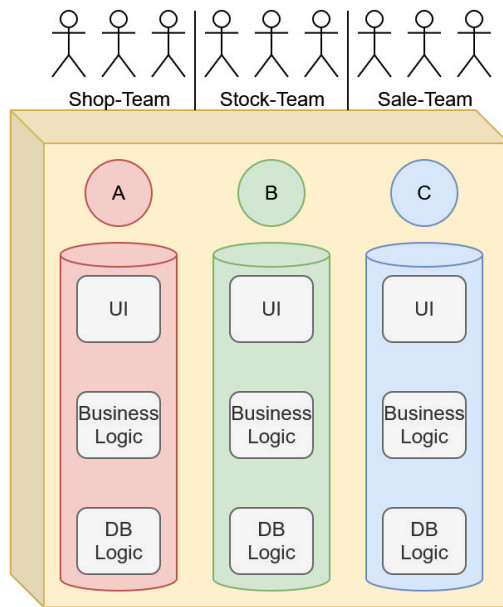


Fig. 2. Illustration of a business application built as microservices. Original illustration from C. Nockemann [3].

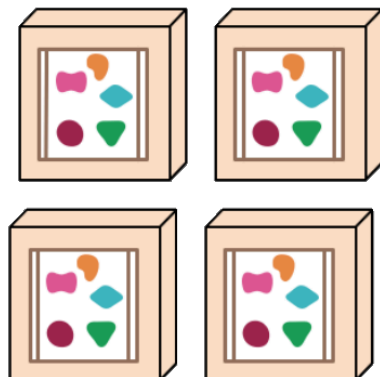
II. PROBLEME / FRAGESTELLUNGEN

Was genau ist Ihre Problemstellung? Welche Fragestellungen behandeln Sie in Ihrer Arbeit?

A monolithic application puts all its functionality into a single process...



... and scales by replicating the monolith on multiple servers



III. METHODISCHES VORGEHEN ZUR ZIELERREICHUNG

Wie gehen Sie bei der Erstellung der Ergebnisse vor (Arbeitsschritte) und welche Methodik praktizieren Sie dabei?

IV. ZIELE UND ANGESTREBTE ERGEBNISSE

Welche Ziele verfolgen Sie mit der Beantwortung der behandelten Fragen und welche wesentlichen Ergebnisse entstehen dabei?

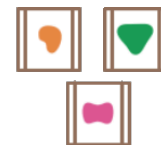
V. WISSENSCHAFTLICHER BEITRAG

Welchen Beitrag zur Bewältigung der Problemstellung leistet Ihre Arbeit? Welche neuen Erkenntnisse können Lesende Ihrer Arbeit erwarten?

REFERENCES

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- [4] M. Daoud, A. E. Mezouari, N. Faci, D. Benslimane, Z. Maamar, and A. E. Fazziki, "Automatic Microservices Identification from a Set of Business Processes," in *Third International Conference on Communications in Computer and Information Science (SADASC 2020)*, M. Hamlich, L. Bellatreche, A. Mondal, and C. Ordóñez, Eds., Cham: Springer International Publishing, 2020, pp. 299–315.
- [5] M. Garriga, "Towards a Taxonomy of Microservices Architectures," in *15th International Conference on Software Engineering and Formal Methods (SEFM 2017)*, A. Cerone and M. Roveri, Eds., Cham: Springer International Publishing, 2018, pp. 203–218.

A microservices architecture puts each element of functionality into a separate service...



... and scales by distributing these services across servers, replicating as needed.

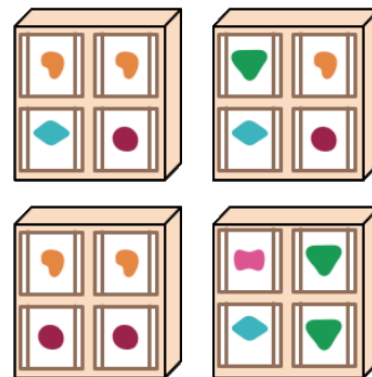


Fig. 3. Scaling and deploying of Monoliths and Microservices by J. Lewis and M. Fowler [2]